



CPIT 370- Project -Phase2

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Your Network Address
192.100.100.0/24

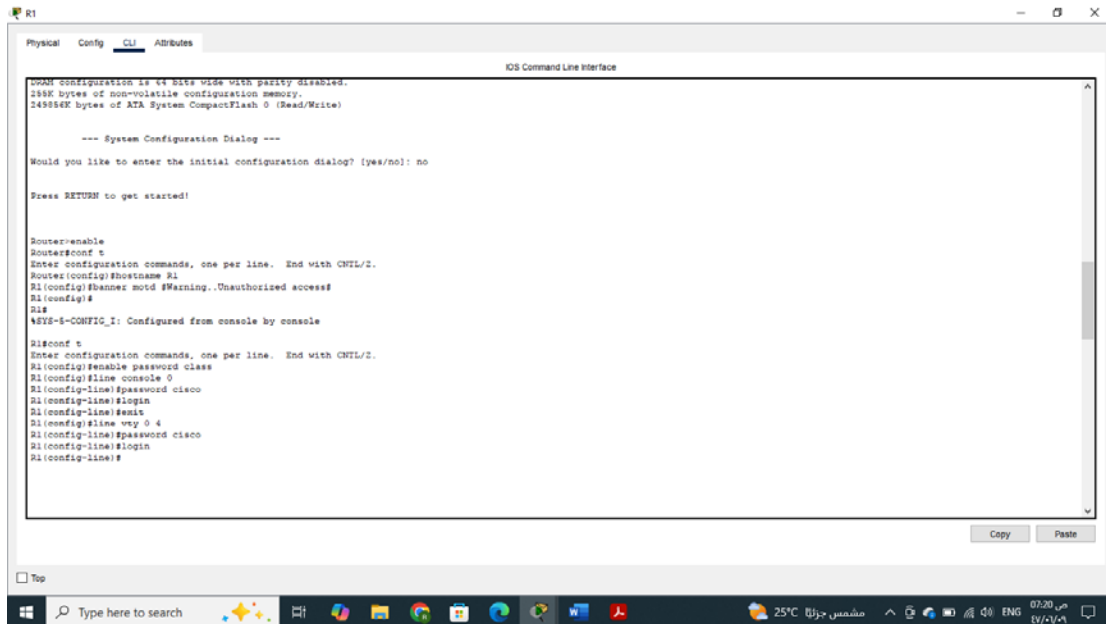
Subnetting Table

Subnet Name	<i>n</i> host bits	Subnet mask (in dot notation & CIDR)	Subnet address (Network address)	First usable IP address	Last usable IP address	Broadcast address
Branch2	6	255.255.255.192 /26	192.100.100.0	192.100.100.1	192.100.100.62	192.100.100.63
Branch1	5	255.255.255.224 /27	192.100.100.64	192.100.100.65	192.100.100.94	192.100.100.95
Branch3	4	255.255.255.240 /28	192.100.100.96	192.100.100.97	192.100.100.110	192.100.100.111
WAN	2	255.255.255.252 /30	192.100.100.112	192.100.100.113	192.100.100.114	192.100.100.115

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	F0/0	192.100.100.65	255.255.255.224	N/A
	F0/1	192.100.100.1	255.255.255.192	N/A
	S0/0/0	192.100.100.113	255.255.255.252	N/A
R2	F0/0	192.100.100.97	255.255.255.240	N/A
	S0/0/0	192.100.100.114	255.255.255.252	N/A
S1	VLAN 1	192.100.100.66	255.255.255.224	192.100.100.65
S2	VLAN 1	192.100.100.2	255.255.255.192	192.100.100.1
S3	VLAN 1	192.100.100.98	255.255.255.240	192.100.100.97
PC1	NIC	192.100.100.3	255.255.255.192	192.100.100.1
Printer1	NIC	192.100.100.4	255.255.255.192	192.100.100.1
PC2	NIC	192.100.100.99	255.255.255.240	192.100.100.97
PC3	NIC	Dynamic	255.255.255.240	192.100.100.97
DHCP	NIC	192.100.100.67	255.255.255.224	192.100.100.65
DNS	NIC	192.100.100.68	255.255.255.224	192.100.100.65
WEB	NIC	192.100.100.69	255.255.255.224	192.100.100.65

R1



The screenshot shows the initial configuration of a Cisco IOS router (R1). The interface displays the following text:

```
ROM Configuration is 64 bits wide with parity disabled.
198K bytes of non-volatile configuration memory.
24904K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

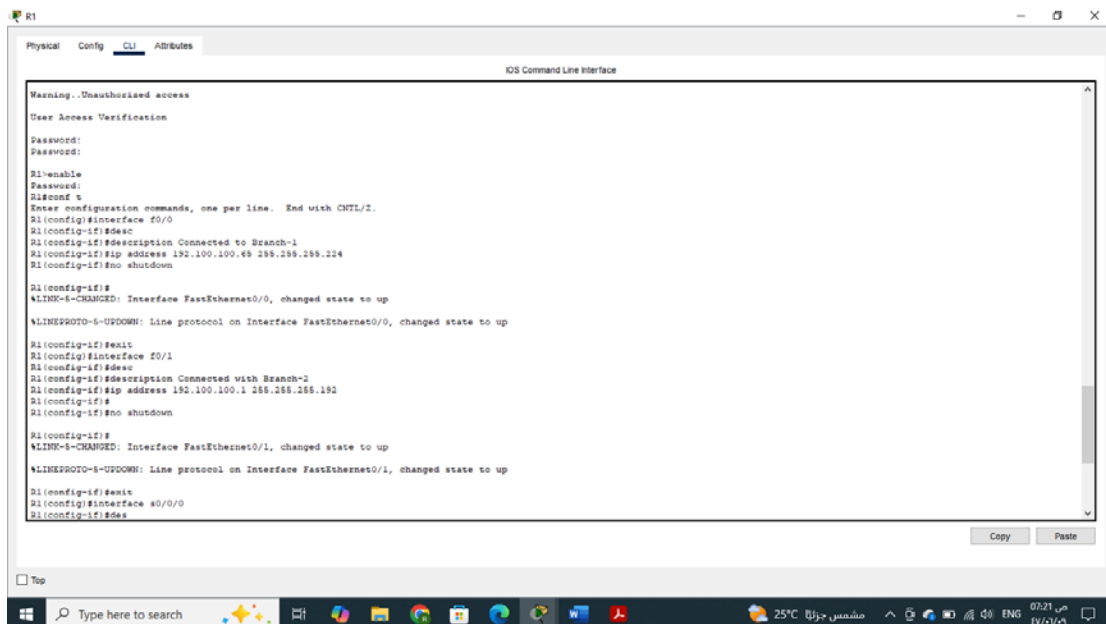
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#banner motd $Warning..Unauthorized access$
R1(config)#
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#enable password class
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#line vty 0 4
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#
```

The Windows taskbar at the bottom shows the date and time as 07/20/2023, 11:11 AM, and the temperature as 25°C.



The screenshot shows the configuration of two interfaces on router R1. The interface displays the following text:

```
Warning..Unauthorized access

User Access Verification

Password:
Password:

R1#enable
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface f0/0
R1(config-if)#desc
R1(config-if)#description Connected to Branch-1
R1(config-if)#ip address 192.100.100.69 255.255.255.224
R1(config-if)#no shutdown

R1(config-if)#
%LINK-6-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#interface f0/1
R1(config-if)#desc
R1(config-if)#description Connected with Branch-2
R1(config-if)#ip address 192.100.100.1 255.255.255.192
R1(config-if)#
R1(config-if)#no shutdown

R1(config-if)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#interface s0/0/0
R1(config-if)#desc
```

The Windows taskbar at the bottom shows the date and time as 07/21/2023, 11:11 AM, and the temperature as 25°C.

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1(config-if)#exit
R1(config)#interface f0/1
R1(config-if)#description
R1(config-if)#description Connected with Branch-2
R1(config-if)#ip address 192.100.100.1 255.255.255.192
R1(config-if)#
R1(config-if)#no shutdown

R1(config-if)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#interface s0/0/0
R1(config-if)#description
R1(config-if)#description Connected with serial to R2
R1(config-if)#ip address 192.100.100.113 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-6-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#exit
R1(config)#
%LINK-6-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R1(config)#
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

Copy Paste

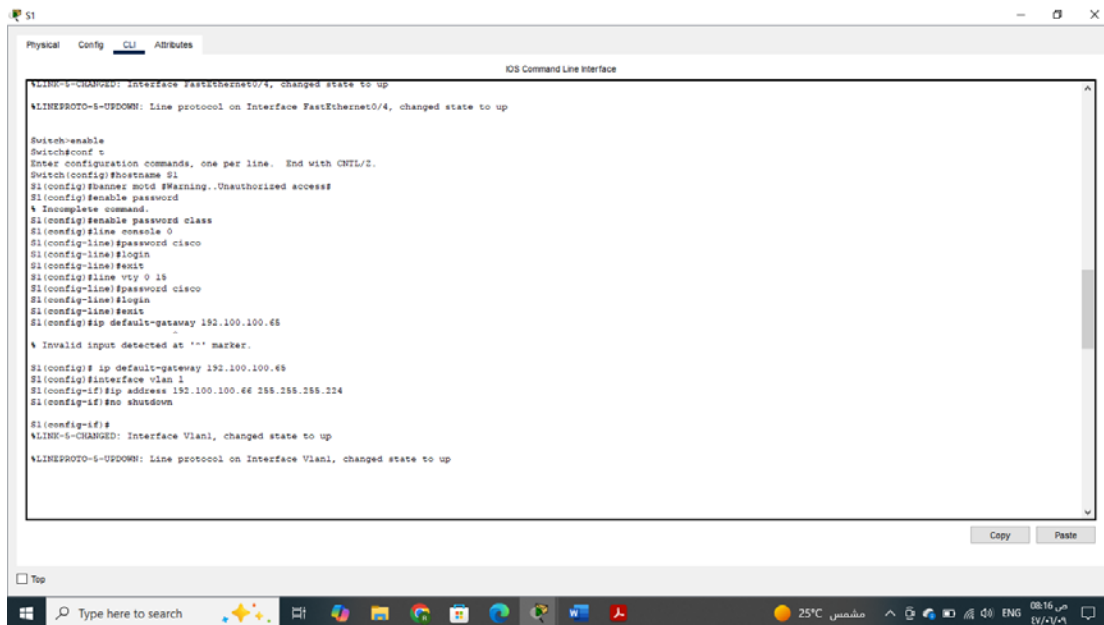
Top

Type here to search

25°C مشمس جزلة

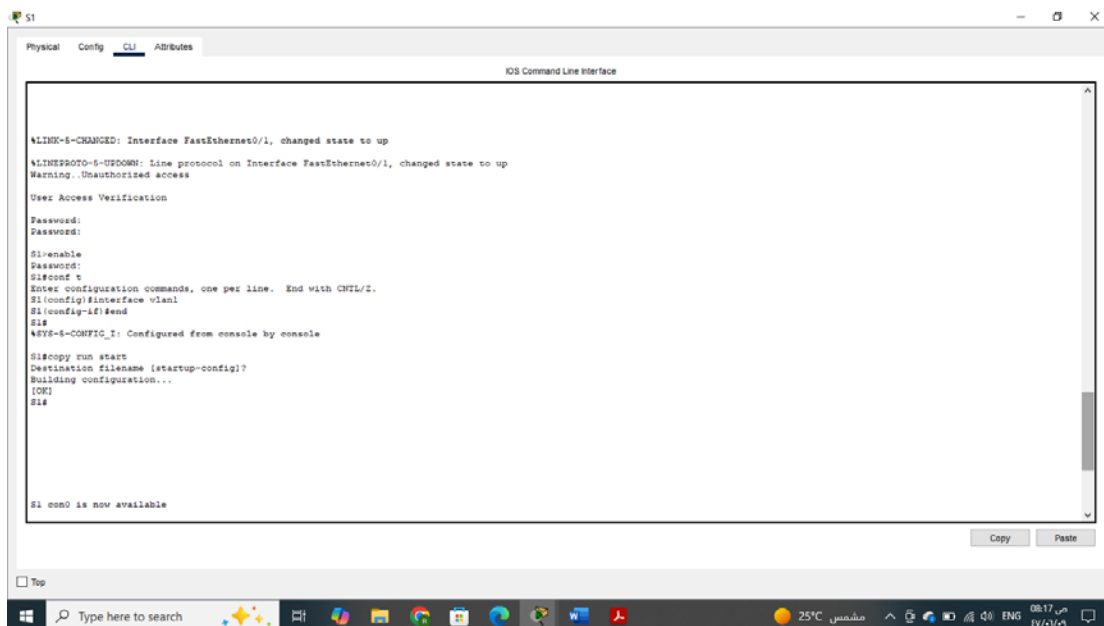
07:21 م 14/1/19

S1



This screenshot shows the configuration of interface FastEthernet0/4 on switch S1. The configuration includes enabling the interface, setting the line protocol to up, and configuring the interface with IP address 192.100.100.66. The configuration is entered in the CLI mode, and the switch is in the configuration mode for the interface.

```
Switch>enable
Switch>conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#banner motd #Warning..Unauthorized access#
S1(config)#enable password
% Incomplete command.
S1(config)#enable password class
S1(config)#line console 0
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#exit
S1(config)#line vty 0 15
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#exit
S1(config)#ip default-gateway 192.100.100.66
% Invalid input detected at '' marker.
S1(config)# ip default-gateway 192.100.100.66
S1(config)#interface Vlan 1
S1(config-if)#ip address 192.100.100.66 255.255.255.224
S1(config-if)#no shutdown
S1(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```



This screenshot shows the configuration of interface FastEthernet0/1 on switch S1. The configuration includes enabling the interface, setting the line protocol to up, and configuring the interface with IP address 192.100.100.66. The configuration is entered in the CLI mode, and the switch is in the configuration mode for the interface.

```
Switch>enable
Switch>conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#banner motd #Warning..Unauthorized access#
S1(config)#enable password
% Incomplete command.
S1(config)#enable password class
S1(config)#line console 0
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#exit
S1(config)#line vty 0 15
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#exit
S1(config)#ip default-gateway 192.100.100.66
% Invalid input detected at '' marker.
S1(config)# ip default-gateway 192.100.100.66
S1(config)#interface Vlan 1
S1(config-if)#ip address 192.100.100.66 255.255.255.224
S1(config-if)#no shutdown
S1(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
Warning..Unauthorized access
User Access Verification
Password:
Password:
S1>enable
Password:
S1>conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#interface vlan1
S1(config-if)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console
S1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
S1#
S1 con0 is now available
```

Task6

The screenshot displays the Cisco Packet Tracer interface. On the left, a network diagram is visible with a 'Server-DNS' connected to a 'Server-PT DHCP' via a switch. A 'Branch-2' section shows a 'PC-PT PC1' connected to a switch. A 'Command Prompt' window is open on the 'PC1' device, showing the results of a ping command to 192.100.100.99. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The command prompt shows two successful ping attempts with detailed statistics.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.100.100.99

Pinging 192.100.100.99 with 32 bytes of data:

Request timed out.
Reply from 192.100.100.99: bytes=32 time=10ms TTL=126
Reply from 192.100.100.99: bytes=32 time=30ms TTL=126
Reply from 192.100.100.99: bytes=32 time=11ms TTL=126

Ping statistics for 192.100.100.99:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 30ms, Average = 17ms

C:\>ping 192.100.100.99

Pinging 192.100.100.99 with 32 bytes of data:

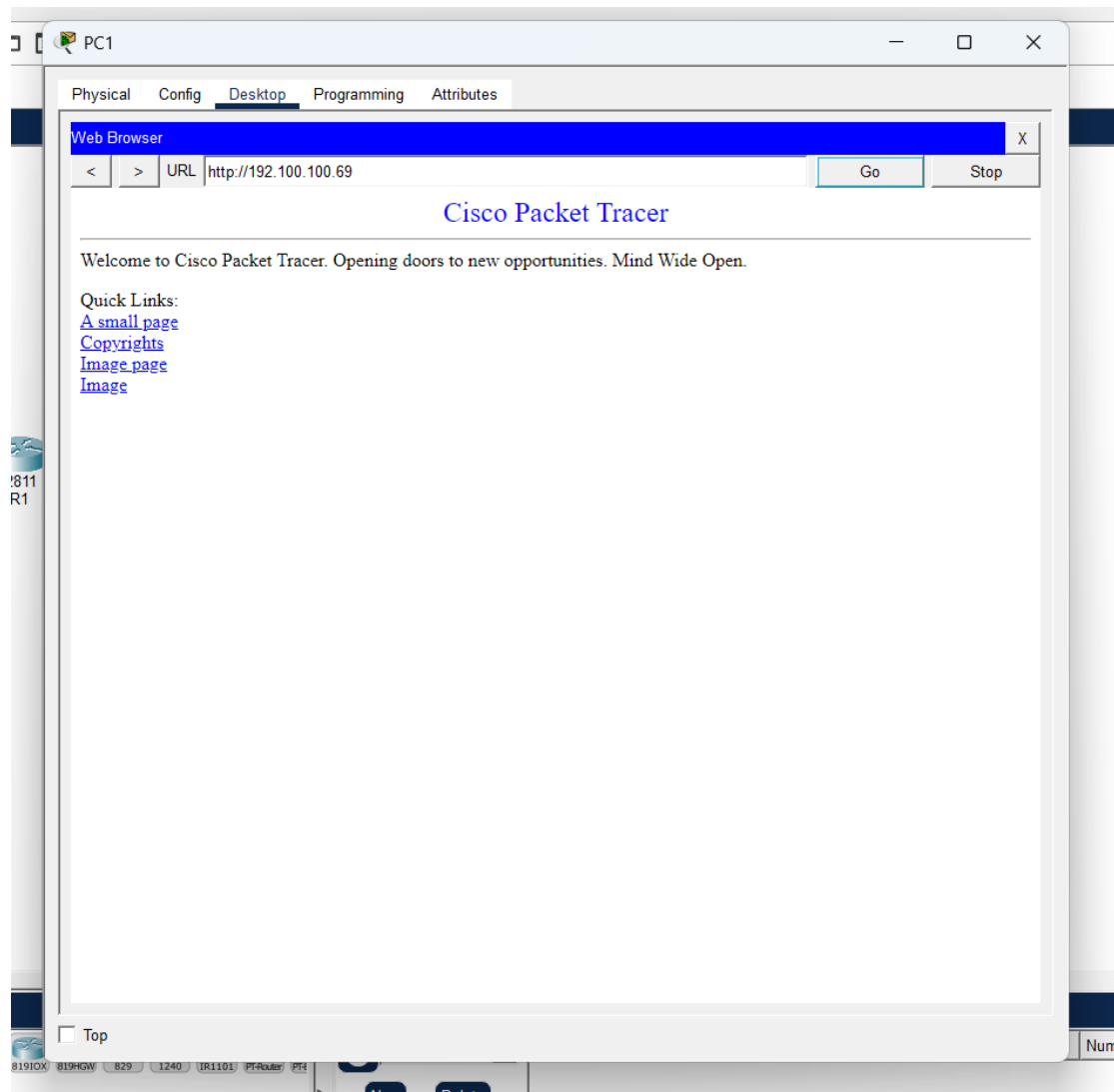
Reply from 192.100.100.99: bytes=32 time=1ms TTL=126
Reply from 192.100.100.99: bytes=32 time=10ms TTL=126
Reply from 192.100.100.99: bytes=32 time=14ms TTL=126
Reply from 192.100.100.99: bytes=32 time=11ms TTL=126

Ping statistics for 192.100.100.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 9ms

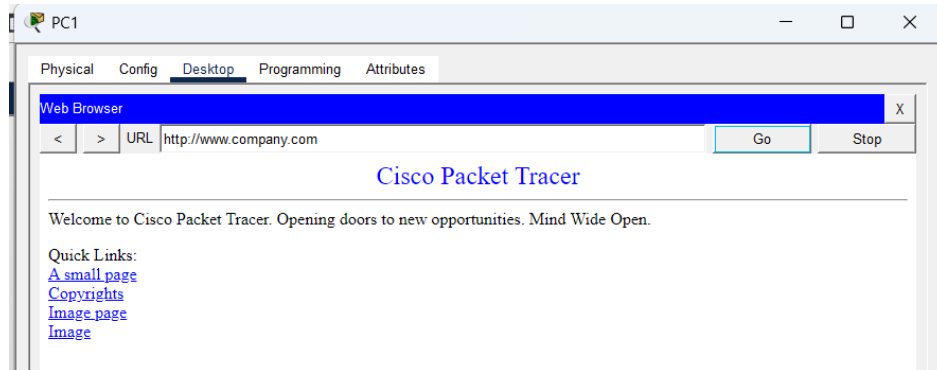
C:\>
```

The bottom of the screen shows the Windows taskbar with the search bar, task view button, and several application icons. The system tray on the right indicates a temperature of 25°C, the date 08/13, and the time 08:13.

Verify the WEB server by typing the website IP address in the browser of PC1.



Verify the ability of client computers to use DNS by browsing the website from PC1 and typing the website name instead of the IP address.



Request DHCP addressing for PC3.

